

Microstructure and the properties of FeCoCuNiSn_x high entropy alloys

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ABSTRACT

FeCoCuNiSn_x high-entropy alloys (*x* denotes the adding the elements amount in atomic percentage) are prepared by an arc furnace. Their microstructure and mechanical properties are investigated. The results show that the alloys have a single FCC solution when Sn content is small, the microstructure of the alloys with increasing Sn content is FCC solution and Cu₈₁Sn₂₂ intermetallic compounds. The alloys possess the high strength and the plasticity. When Sn content is between 0.05 and 0.07, the maximum elongation strain and the maximum tensile strength can reach 19.8% and 633 MPa, respectively. The adding of Sn leads to the increase of tensile strength.

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1. Introduction

The general strategy for developing alloys is to select one or two elements as principal components for primary properties and other minor elements for the acquisition of definite microstructure and properties. It is known that when the principal element has a high solubility; the strength of the alloys can increase due to the solution hardening. If the solution structure is kept, the alloy would have a better combination of strength and plastics. As the amount of alloying element increases, intermetallic compounds, however, are typically formed in the alloys due to the limited solid solubility, which results in enhanced strength at the expense of accompanying embrittlement. To realize the goal of a solution structure, a new alloy design concept, the high entropy (HE) alloy was explored by Yeh et al. based on the thermodynamic consideration, which consists of a simple FCC or BCC structure [1–5]. Following Boltzmann's hypothesis on the relationship of entropy complexity [2], the configurational entropy change, ΔS_{conf} , during the formation of a solid solution from *n* elements with equimolar fractions, may be calculated from the following equation:

$$\Delta S_{\text{conf}} = -R \ln \left(\frac{1}{n} \right) = R \ln(n) \quad (1)$$

where *R* is the gas constant. When *n*=5, ΔS_{conf} =1.61*R*, which approaches the magnitude of the melting entropy of the most intermetallic compounds (about *R*-2*R*). As results, intermetallic compounds are absent and a single solid solution is built with

structures of a single FCC, a single BCC or FCC+BCC [2–5]. Due to the above structural and compositional characteristics, the alloys are ductile and working-hardenable, and have high strength [3–6]. Thus, the alloys have potential applications as high-temperature structure materials and work tools. Also, due to these possible applications, the properties of the alloys, such as strength, hardness, wear resistance, were investigated [5–15]. To improve the properties of the alloys, the effects of alloying of B, Cr, Ti, Fe, Al and V elements on the structure and properties have been considered [7–17].

As everyone knows, Sn has a low melting point of 505.06 K, but a high boiling point of 2876 K. Differing from other elements, there is a wide difference of about 2400 K between the two temperatures. Sn is very soft at room temperature, and has low solubility with other elements, therefore, when a small quantity of Sn adds in to the alloys, it will be segregated to grain boundary, accordingly improve the plastic property of the high entropy alloy [18].

In this contribution, FeCoCuNiSn_x alloys with different Sn contents are fabricated. The corresponding microstructures and mechanical properties are measured, the effects of Sn content on the alloy microstructure and tensile properties are investigated.

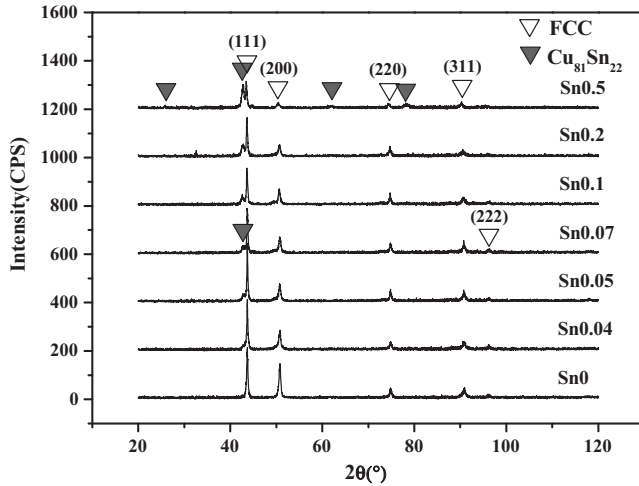
2. Experimental methods

FeCoCuNiSn_x alloys are prepared by a vacuum furnace under argon atmosphere. The purity of all the raw elemental metal is above 99.9%. The rod ingot of the alloy with a diameter of 6 mm is obtained in a cold copper hearth. The corresponding cooling rate is about 10³ K/s. Table 1 shows the composition of the alloys. The phase structure analysis of the samples was made by Rigaku D/max 2500 X-ray diffractometer at 50 kV and 250 mA. Scanning

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Table 1
the composition of the alloys used in the experiment (at.%).

Alloy No.	Fe	Co	Cu	Ni	Sn	Atomic ratio
1	25.0000	25.0000	25.0000	25.0000	0	FeCoCuNi
2	24.8756	24.8756	24.8756	24.8756	0.4975	FeCoCuNiSn _{0.02}
3	24.7525	24.7525	24.7525	24.7525	0.9901	FeCoCuNiSn _{0.04}
4	24.6914	24.6914	24.6914	24.6914	1.2346	FeCoCuNiSn _{0.05}
5	24.5700	24.5700	24.5700	24.5700	1.7199	FeCoCuNiSn _{0.07}
6	24.3902	24.3902	24.3902	24.3902	2.4390	FeCoCuNiSn _{0.1}
7	23.8095	23.8095	23.8095	23.8095	4.7619	FeCoCuNiSn _{0.2}
8	22.2222	22.2222	22.2222	22.2222	11.1111	FeCoCuNiSn _{0.5}

**Fig. 1.** X-ray diffraction curve analysis of the alloys with different Sn contents.

angles ranged from 20 to 120° with a scanning rate of 5°/min. The microstructure of the alloys was observed by scanning electron microscope (SEM) and transmission electron microscope (TEM), respectively. The selected area composition analysis of the samples was carried out by an energy dispersive spectrometry (EDS). The tensile stress–strain curves of the alloy were measured by an 810 Material Test System (MTS) at a crosshead rate with 0.2 mm/min.

3. Results and discussion

Fig. 1 shows X-ray analysis results of the alloys with different Sn contents. When $x=0$, the alloy is a single FCC crystal structure. $\text{Cu}_{81}\text{Sn}_{22}$ phase appears with increasing of Sn contents. Because $\text{Cu}_{81}\text{Sn}_{22}$ is hard and brittle phase, which will decrease the tensile strength and the toughness of the alloys. **Fig. 2** shows the lattice constant of the alloys with different Sn contents. It can find that the lattice constant increases with the increasing Sn contents from 0.3586 nm to 0.3606 nm because Sn is larger than the other elements in the alloys. The adding of Sn content not only introduces the lattice strain, but also affects the microstructure of the alloys.

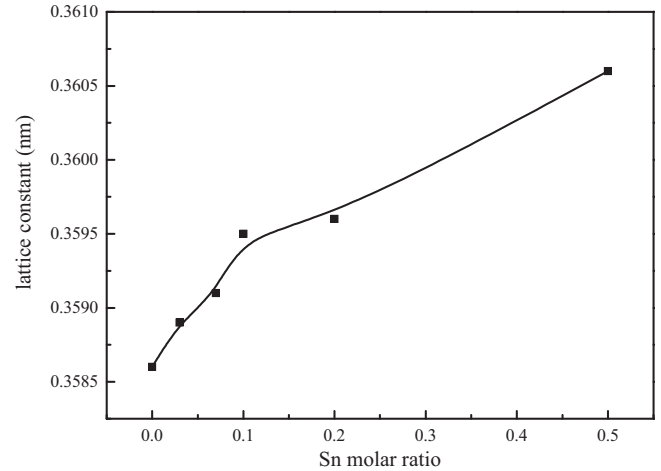
**Fig. 2.** Lattice constant curve of the alloys with different Sn contents.

Fig. 3 is the microstructure of the alloys with different Sn contents. The results show that the structure of the alloys is a typical dendritic crystal structure. As Sn content increases, the morphology of the microstructure varies and the grains coarsen. **Table 2** is the element distribution in the alloys determined by EDS. The result shows that the dendritic regions are rich Fe and Co, interdendritic region is rich Cu and Sn, the distribution of Ni is more uniform in dendritic region and interdendritic region. When Sn content is less than 0.07, the Sn element is almost in interdendritic region that can improve the plastic property of the alloys. As Sn unceasingly increases, redundant Sn will combine with Cu to form intermetallic compounds, which will seriously depress the mechanical properties of the alloys.

Fig. 4 is the TEM micrographs of FeCoCuNiSn_{0.2} alloy and FeCoCuNiSn_{0.05} alloy, respectively. It can find that there are massive precipitates in FeCoCuNiSn_{0.2} alloy, which are mostly $\text{Cu}_{81}\text{Sn}_{22}$ intermetallic compounds determined from the diffraction spots. Because $\text{Cu}_{81}\text{Sn}_{22}$ phase is hard and brittle phase, the tensile strength of the alloy can be decreased. However, it does not be found that the intermetallic compounds are precipitated in FeCoCuNiSn_{0.05} alloy. It can clearly observe the twin crystals and

Table 2
Dendritic (De) and interdendritic (In) compositions determined by EDS (at.%).

Alloy	Fe		Co		Ni		Cu		Sn	
	De	In	De	In	De	In	De	In	De	In
$x=0$	30.80	16.01	28.01	11.94	25.83	19.98	15.35	52.07	–	–
$x=0.02$	30.26	15.05	27.50	11.21	25.63	20.79	16.48	51.75	0.13	1.20
$x=0.04$	28.23	14.26	30.31	12.45	22.96	21.68	18.36	50.23	0.14	1.38
$x=0.05$	28.82	16.58	29.85	17.82	24.07	24.51	17.12	39.61	0.14	1.47
$x=0.07$	33.09	13.47	24.23	11.30	25.27	25.12	22.24	42.23	0.17	2.88
$x=0.1$	27.08	15.55	29.02	8.54	23.57	23.38	19.25	41.16	1.08	11.37
$x=0.2$	29.09	6.12	27.49	6.46	23.43	21.71	17.22	45.59	2.77	20.12
$x=0.5$	31.51	5.23	29.76	7.18	20.44	20.43	13.21	46.45	5.08	20.71

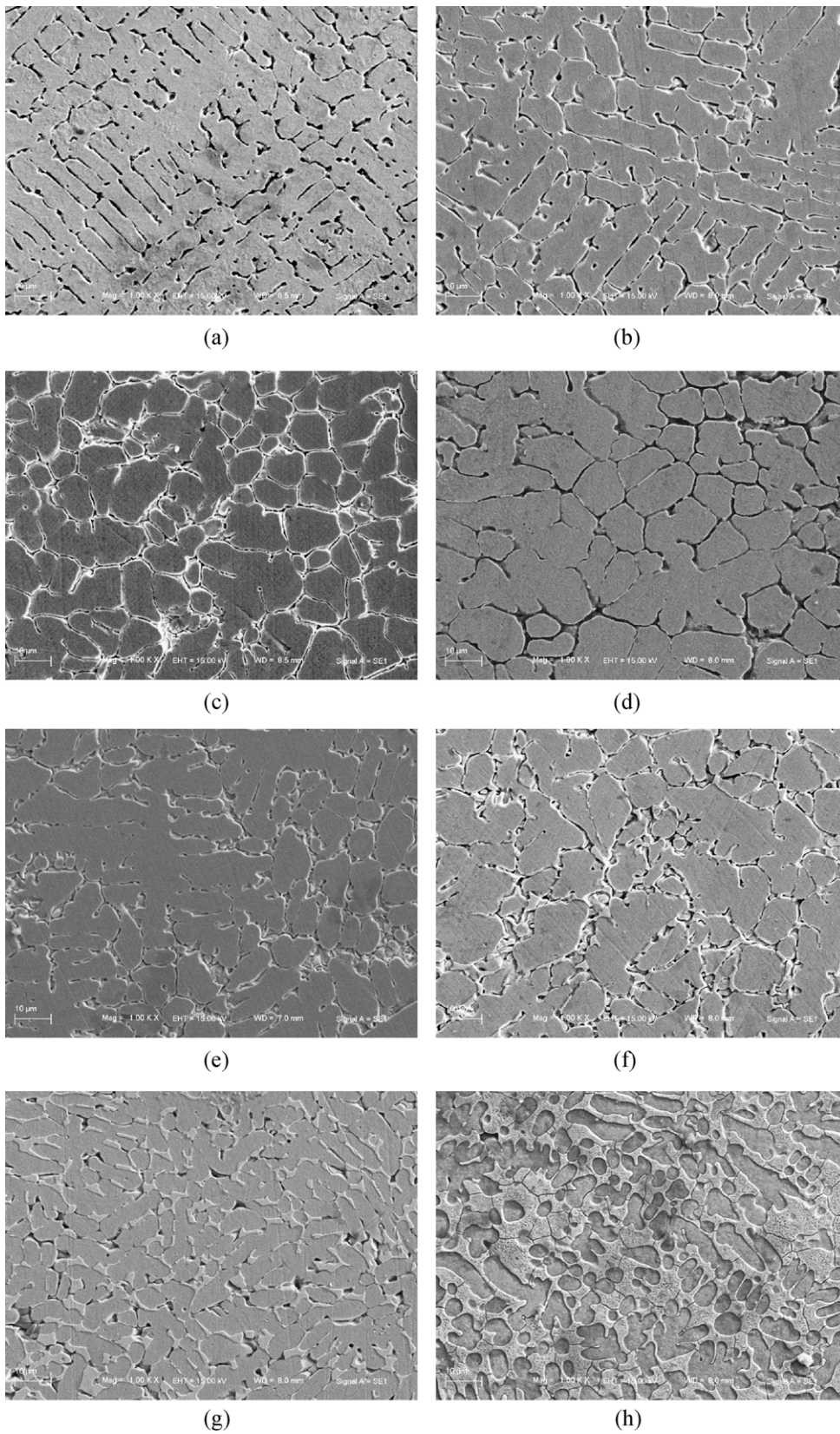


Fig. 3. SEM micrograph of the alloys with different Sn contents: (a) FeCoCuNi alloy, (b) FeCoCuNiSn_{0.02} alloy, (c) FeCoCuNiSn_{0.04} alloy, (d) FeCoCuNiSn_{0.05} alloy, (e) FeCoCuNiSn_{0.07} alloy, (f) FeCoCuNiSn_{0.1} alloy, (g) FeCoCuNiSn_{0.2} alloy and (h) FeCoCuNiSn_{0.5} alloy.

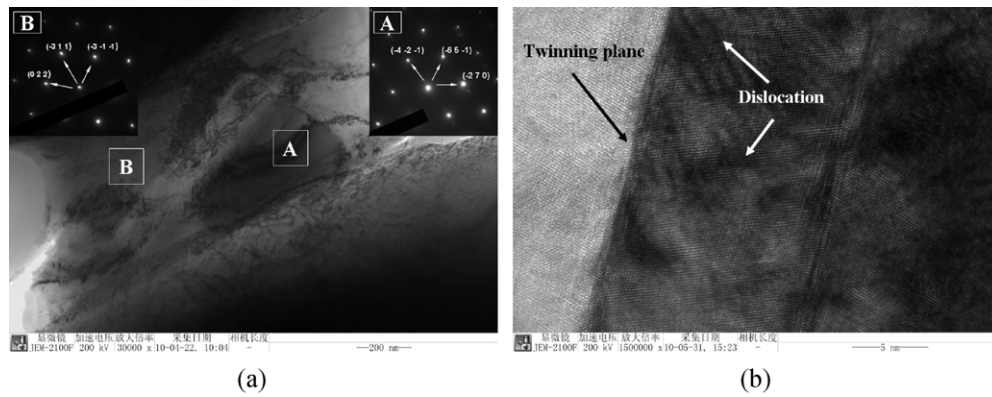


Fig. 4. TEM diffraction measurement and the diffraction spot of (a) FeCoCuNiSn_{0.2} alloy. Twin crystals and dislocations of (b) FeCoCuNiSn_{0.2} alloy.

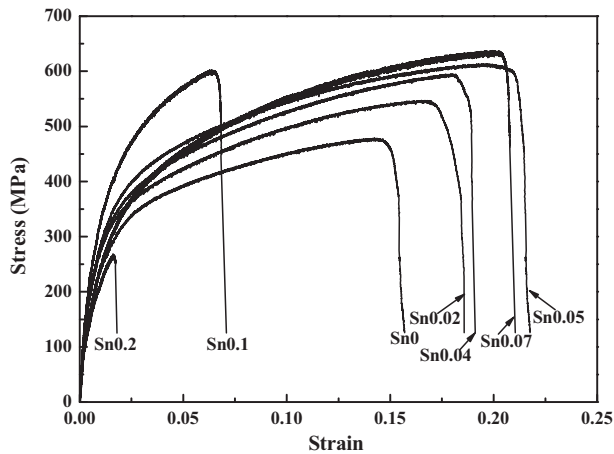


Fig. 5. Tensile strain and tensile strength curves of alloys with different Sn contents.

lots of dislocations, which is good for mechanical properties of alloy [19–21].

Fig. 5 is the stress–strain curves of the FeCoCuNiSn_x alloys with different Sn contents. The results show that the largest elongation strain for FeCoCuNiSn_{0.05} alloy is 19.8%, and the highest tensile strength for FeCoCuNiSn_{0.07} alloy is 633 MPa. FeCoCuNiSn_{0.2} alloy have the lowest elongation strain and the tensile strength, just 1.2% and 263 MPa. Because the brittleness of FeCoCuNiSn_{0.5} alloy is too large, its tensile strength cannot be measured. Fig. 6 shows that the strain and stress change regularly with the change of Sn content. When Sn content is less than 0.07, the elongation strain and the tensile strength of FeCoCuNiSn_x increase with increasing Sn content,

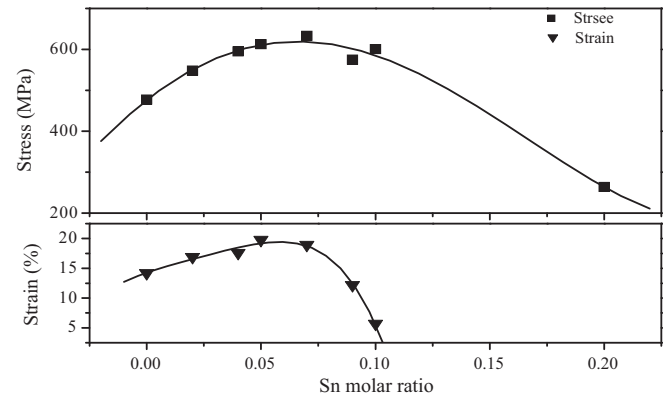


Fig. 6. The stress–strain curves of the FeCoCuNiSn_x alloys.

however, Sn content is more than 0.07, the tendency is opposite, and the elongation strain changes more tempestuously than that of the tensile strength. Fig. 7 is the fracture morphology of the tensile samples of FeCoCuNiSn_{0.05} alloy and FeCoCuNiSn_{0.2} alloy, respectively. It can be observed ductile fracture feature for FeCoCuNiSn_{0.05} alloy and the cleavage fracture feature for FeCoCuNiSn_{0.2} alloy, which is agree to the results of the tensile experiment.

4. Summary

FeCoCuNiSn_x high entropy alloys were prepared with a FCC crystal structure when Sn content is less than 0.05, Cu₈₁Sn₂₂ phase appears when Sn content is more than 0.05. The dendritic area is rich Fe and Co and the interdendritic area is rich Cu and Sn, the

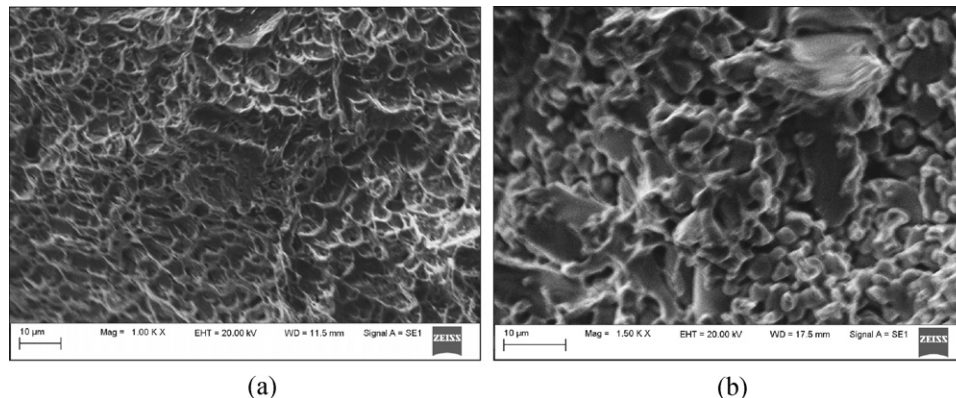


Fig. 7. SEM fracture morphology of the alloys: (a) FeCoCuNiSn_{0.05} alloy and (b) FeCoCuNiSn_{0.2} alloy.

distribution of Ni is more uniform in dendritic region and interdendritic region. When Sn content is between 0.05 and 0.07, the alloys possess the high strength and the plasticity, the maximum elongation strain and the maximum tensile strength can reach 19.8% and 633 MPa, respectively.

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